Paper_Se_Sp_analysis

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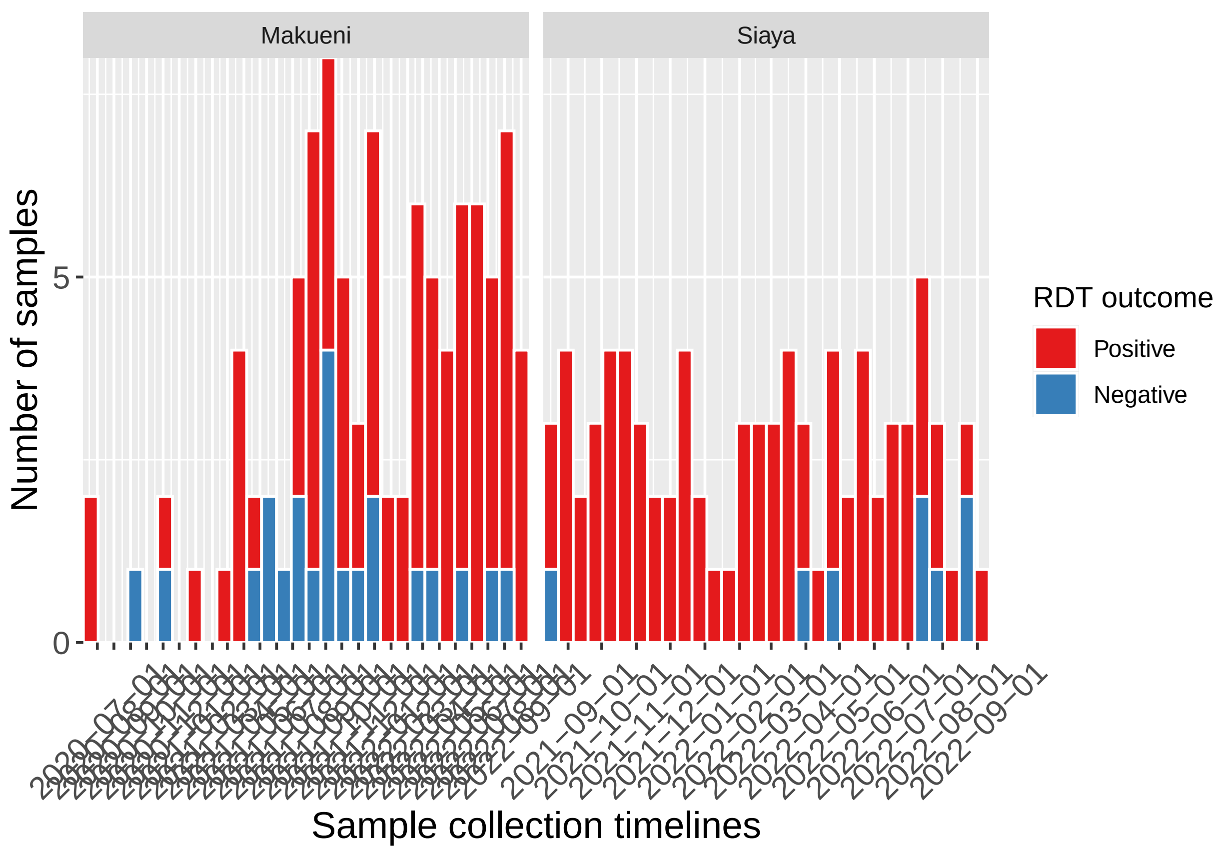
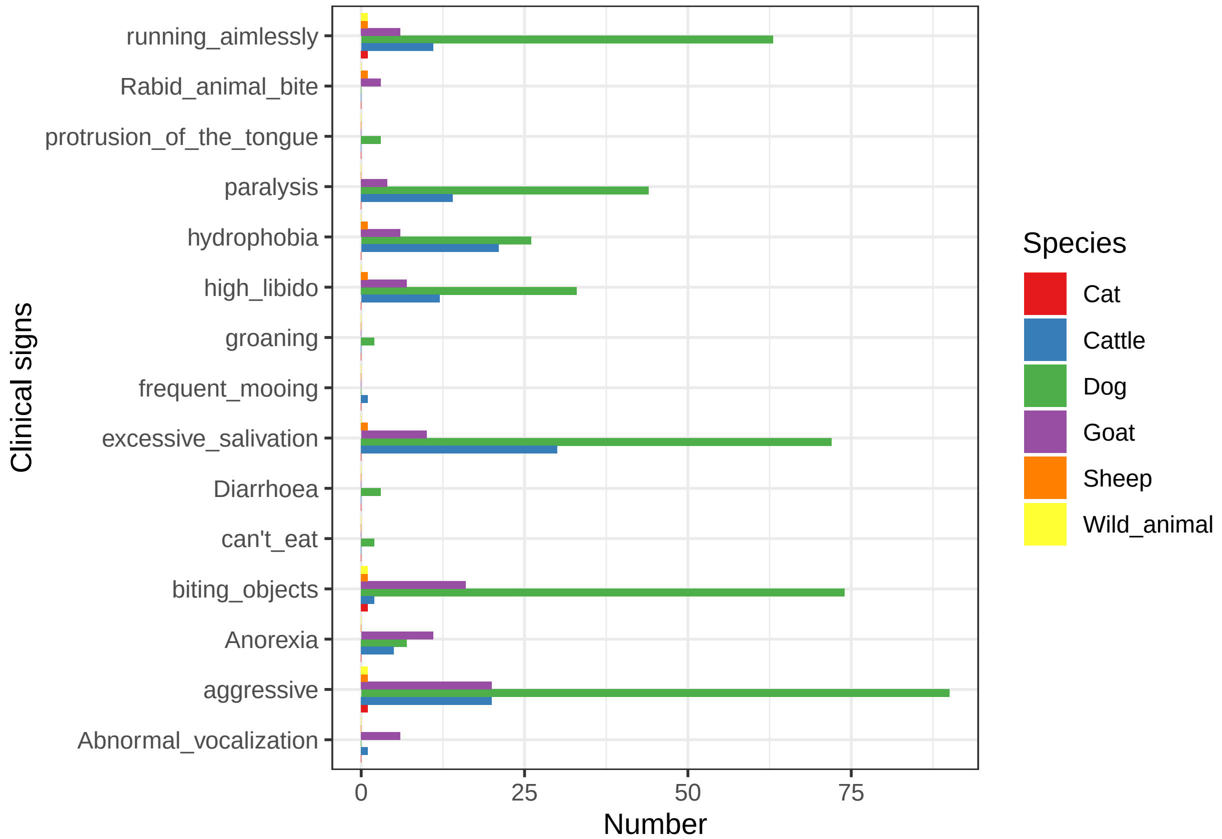
Time series



clinical signs

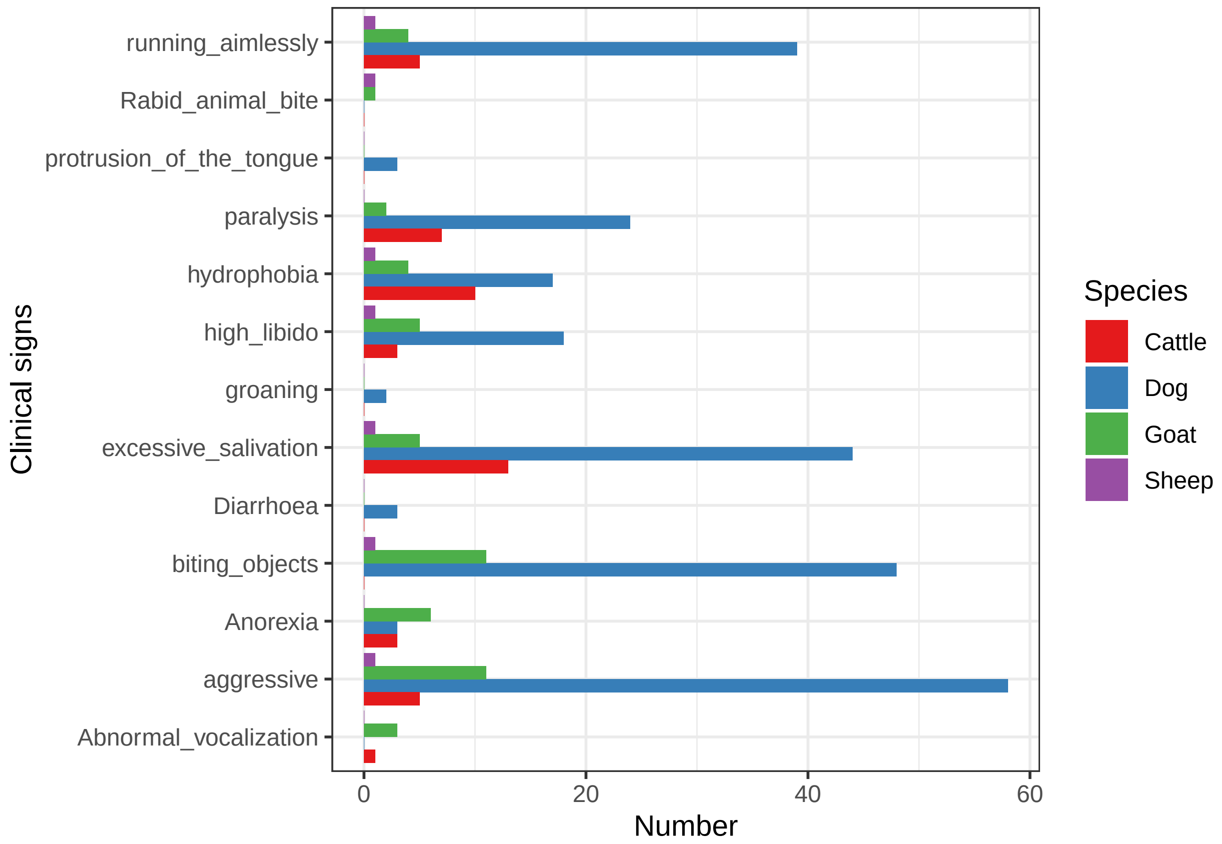
```
# cleaning the contact tracing data

ct_data <- import(here::here("data", "clean", "ct_data.rds"))

# figure using the entire dataset (205 samples)
```

```r
Fig <- ct_data |>
  select(sample_collection_dt, species, cnty_name, clinical_signs_, lab_diagnostics_pcr, rapid_diagnosti
  #filter(!lab_diagnostics_pcr %in% "") |>
  group_by(species, clinical_signs_) |>
  dplyr::count()

Fig_new = Fig
for(i in unique(Fig$clinical_signs_)){
  for(j in unique(Fig$species)){
    if(nrow(Fig[Fig$clinical_signs_ == i &
                    Fig$species == j,]) == 0){
      new_row = data.frame(
                  clinical_signs_ = i,
                  species = j,
                  n = 0)
      Fig_new = rbind(Fig_new, new_row)
    }
  }
}

ggplot(data = Fig_new)+
  geom_col(aes(y=n, x = clinical_signs_,fill = species), position = "dodge") +
  labs(x = "Clinical signs", y = "Number", fill="Species")+
  scale_fill_brewer(palette = "Set1", direction = 1)+
  coord_flip()+
  theme_bw()
```




```
ggsave("/Users/mumbuamutunga/Library/CloudStorage/Dropbox/Desktop/Sp:Se_Analysis/Sp:Se_Manuscript/Figure


# figure using the sequenced samples

Fig_seq <- ct_data |>
  select(sample_collection_dt, species, cnty_name, clinical_signs_, lab_diagnostics_pcr, rapid_diagnosti
  filter(!lab_diagnostics_pcr %in% "") |>
  group_by(species, clinical_signs_) |>
  dplyr::count()

Fig_new_seq <- Fig_seq

for(i in unique(Fig_seq$clinical_signs_)){
  for(j in unique(Fig_seq$species)){
    if(nrow(Fig_seq[Fig_seq$clinical_signs_ == i &
                     Fig_seq$species == j,]) == 0){
      new_row = data.frame(
                  clinical_signs_ = i,
                  species = j,
                  n = 0)
      Fig_new_seq = rbind(Fig_new_seq, new_row)
    }
  }
}
```

```r
ggplot(data = Fig_new_seq)+
  geom_col(aes(y=n, x = clinical_signs_,fill = species), position = "dodge") +
  labs(x = "Clinical signs", y = "Number", fill="Species")+
  scale_fill_brewer(palette = "Set1", direction = 1)+
  coord_flip()+
  theme_bw()
```




```r
ggsave("/Users/mumbuamutunga/Library/CloudStorage/Dropbox/Desktop/Sp:Se_Analysis/Sp:Se_Manuscript/Figur

#incidence of positive cases per year

inc_mak <- combined |>
  select(date_collected, county, rapid_diagnostic_test) |>
  filter(rapid_diagnostic_test %in% "Positive") |>
  filter(county %in% "Makueni") |>
  mutate(year = format(date_collected, "%Y")) |>  # Extract the year
  group_by(year) |>                          # Group by year
  count()


inc_siaya <- combined |>
  filter(rapid_diagnostic_test %in% "Positive") |>
  filter(county %in% "Siaya") |>
  mutate(year = format(date_collected, "%Y")) |>  # Extract the year
  group_by(year) |>                          # Group by year
```

```
  count()

# if in two counties we have 94 samples that are positive if we could escalate this to the remaining 45
```

ct_data_ <- import(here::here("data", "clean", "ct_data_.rds"))

ct_plot_dt <- import(here::here("data", "clean", "ct_plot_dt.rds"))

sample_data <- import(here::here("data","clean","sample_data_clean.rds"))

mapping_data <- import(here::here("data","clean","ct_data1.rds"))

makueni_sd <-sample_data |> filter(cnty_name %in% "Makueni") |> filter(!Latitude %in% c("-1.3950553815811872"))

projcrs <- "+proj=longlat +datum=WGS84 +no_defs +ellps=WGS84 +towgs84=0,0,0"

makueni_sd <- st_as_sf(x = makueni_sd,
coords = c("Longitude", "Latitude"), crs = projcrs)

siaya_sd <-sample_data |> filter(cnty_name %in% "Siaya")

siaya_sd <- st_as_sf(x = siaya_sd,
coords = c("Longitude", "Latitude"), crs = projcrs)